Proposed Exercises—Answers and Solutions

Chapter 1

1.3.1 (b)	1.3.2 (a)
1.3.3 (a)	1.3.4 (d)
1.3.5 (c)	1.3.6 (c)
1.3.7 (b)	1.3.8 (a)
1.3.9 (a)	1.3.10 (d)
1.3.11 (a)	1.3.12 (a)
1.3.13 (c)	1.3.14 (d)
1.3.15 (c)	1.3.16 (b)
1.3.17 (b)	1.3.18 (a)
1.3.19 (b)	1.3.20 (c)
1.3.21 (b)	1.3.22 (b)
1.3.23 (c)	1.3.24 (c)

1.3.25 $n = 2$

$$\dim(x) = \frac{\dim(v^n)}{\dim(a)}$$
$$\dim(x) = L$$
$$\dim(v) = LT^{-1} \therefore \dim(v^n) = \left(LT^{-1}\right)^n$$
$$\dim(a) = LT^{-2}$$
$$L = \frac{\left(LT^{-1}\right)^n}{LT^{-2}}$$

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P. P. Novellino do Rosario, A. Mendes, *Metrology and Measurement Uncertainty*,
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1.3.26 (c)
1.3.27 kg s^{-3}

$$I = \frac{E}{S \cdot t}$$

Unit of E: m^2 kg s^{-2}
Unit of S: m^2
Unit of t: s
Unit of I: $\frac{m^2 \text{ kg } s^{-2}}{m^2 \text{ s}} = \text{kg } s^{-3}$

1.3.28 (d)
1.3.29 kg m^{-1}

$$k = \frac{f}{v^2}$$

Unit of f: m kg s^{-2}
Unit of v^2: m^2 s^{-2}
Unit of k: $\frac{m \text{ kg } s^{-2}}{m^2 s^{-2}} = \text{kg } m^{-1}$

Chapter 2

2.11.1 I'm afraid I have to disagree. Traceability is only guaranteed when the instrument, even new, is calibrated with recognized standards and accepted nationally or internationally.
2.11.2 Metrology is the science of measurement and its applications and includes all theoretical and practical aspects of measurement, whatever the uncertainty of measurement and the field of application.
2.11.3

Legal Metrology
It is the area of metrology closest to the ordinary citizen, whose primary function is to protect products and services that involve and need some measurement. It is defined by the International Organization of Legal Metrology (OIML) as: "the application of legal requirements for measurement and instruments." Metrological regulations based on the OIML guidelines establish the technical requirements, metrological control, use, and marking requirements, as well as the requirements of the units of measure that must be met by manufacturers and by users of the measuring instruments.

In addition to commercial activities, measuring instruments used in official activities, medical areas, medicine manufacture, occupational, environmental

standard, according to the guidelines established by the International Laboratory Accreditation Cooperation (ILAC) and the Good Practices Codes (GPC) from the Organization for Economic Co-Operation and Development (OECD).

2.11.9

A technical standard is a document established by consensus and approved by a recognized organism that provides minimum rules, guidelines, or characteristics for activities or their results, aiming to obtain a great degree of sorting in a given context. The technical standard is voluntary, that is, not mandatory by law.

A technical regulation is a document adopted by an authority with legal power that contains mandatory rules and establishes technical requirements, either directly by reference to technical standards or by incorporating their content, in whole or in part. In general, technical regulations aim to ensure aspects related to health, safety, environment, consumer protection, and fair competition. Compliance with a technical regulation is mandatory, and non-compliance with the corresponding punishment is illegal.

2.11.10 VIM is a document that seeks international harmonization of the terminologies and definitions used in metrology and instrumentation.

Chapter 3

3.3.1

(a) 34.4 m	(b) 23.9 m
(c) 8.4 m	(d) 19.7 m
(e) 43.5 m	(f) 43.9 m
(g) 52.4 m	(h) 66.7 m

3.3.2

(a) 4	(b) 2
(c) 1	(d) 4

3.3.3

(a) 479 m	(b) 642 kg
(c) 123 L	(d) 56.2 cm

3.3.4

(a) 89.5 m
(b) 8.2 m^2
(c) 5.55 m

Chapter 4

4.5.1

(a) 0.1 bar
(b) 0.05 bar
(c) 0.30 bar

4.5.2

(a) 2 °C
(b) 1 °C
(c) 20 °C

4.5.3 11 bar
4.5.4 (b)
4.5.5 (b)
4.5.6

(a) Maximum error = 0.8/200 = 0.4%; Hysteresis = 1.2/200 = 0.6 %
(b) Maximum error = 0.8/65 = 1.23%; Hysteresis = 1.2/65 = 1.85 %

4.5.7

(a) $\overline{x} = \frac{\sum x_i}{10} = 15.97\ \Omega$
(b) $B = 15.97 - 15.977 = -0.007\ \Omega = -0.01\ \Omega$
(c) $E = (15.95 - 15.977)\ \Omega = -0.027\ \Omega = -0.03\ \Omega$

4.5.8

(a) $B = (20.5 - 20.0)$ °C = 0.5 °C
(b) $E = (21.0 - 20.0)$ °C = 1 °C

4.5.9 C = (45 – 1) psi = 44 psi
4.5.10 (a)
4.5.11 (b)
4.5.12 (d)
4.5.13 (d)
4.5.14

(a) 1 °C; 2 °F
(b) 1 °C; 2 °F
(c) 21 °C; 70 °F

4.5.15

(a) 0.5 V
(b) 0.25 V
(c) (0 a 15) V
(d) 2% × 15 V = 0.3 V

4.5.16

(a) The precision of an instrument is related to little or no dispersion of the measured values. This can be found in the standard deviation of measurements. The standard deviation of multimeter one is 0.01 Ω, and multimeter two is 0.02 Ω. Therefore, multimeter 1 is more precise.
(b) The accuracy of a measurement instrument is related to its lower error; multimeter one is more accurate since its error is 0.03 Ω, and the multimeter two error equals 0.04 Ω.

4.5.17

(a) Point 5, the bias is zero.
(b) Point 2, the highest bias, 0.004 g.
(c) Mean = 5.004 g
Bias in this point = 0.002 g
Correct value = (5.004 – 0.002) g = 5.002 g

4.5.18

(a) Correction of the standard = (50.2 + 0.3) °C = 50.5 °C
(b) Mean of the object = 50 °C
Bias of the object = (50 – 50.5) °C = − 0.5 °C = 0 °C (the resolution of the object is 1 °C)
(c) Correction = 0 °C.

Chapter 5

5.11.1

(a) 2 km/h
(b) 2 %
(c) 4 %
(d) 40 %
(e) The point of 200 km/h = 1 %.

5.11.10

(a) $\bar{x}_{\text{corrected}} = \bar{x} - B = 100.0035 - (-0.0050) = 100.0085$ g
(b) $u_{\text{repeat}} = 0.000176383$ g
(c) $u_c = \sqrt{u_{\text{repeat}}^2 + u_{\text{scale}}^2} = \sqrt{0.000176383^2 + 0.0004^2} = 0.000437163$ g

$$v_{\text{eff}} = \frac{u_c^4}{\frac{u_A^4}{2}} = 75,5 \rightarrow k = 2.03 \rightarrow U = 2.03 \times 0.000437163 \text{ g} = 0.0009 \text{ g}$$

5.11.11

(a) $\bar{x}_{\text{corrected}} = \bar{x} - B = 12.005 - 0.0015 = 11.9990$ g
(b) $u_{\text{repeat}} = 0.00006667$ g
(c) (c) B = 0.0015 g
(d) $u_c = \sqrt{u_{\text{repeat}}^2 + u_{\text{scale}}^2} = \sqrt{0.00006667^2 + 0.00014218^2} = 0.000157034$ g

$$v_{\text{eff}} = \frac{u_c^4}{\frac{u_A^4}{2} + \frac{u_{\text{scale}}^4}{25}} = 17,93 \rightarrow k = 2.16 \rightarrow U = 2.16 \times 0.000157034 \text{ g} = 0.0003 \text{ g}$$

5.11.12 (c)

Chapter 6

6.6.1

$$M = (1\ 000 \pm 1)\ \text{g} \quad k = 2.00 \text{ and } 95.45\ \%. \quad u_M = 1\,\text{g}/2 = 0.5\ \text{g}$$

$$D = (8.000 \pm 0.002)\ \text{cm} \quad k = 2.00 \text{ and } 95.45\%. \quad \text{u}_\text{D} = 0.002\ \text{cm}/2 = 0.001\ \text{cm}$$

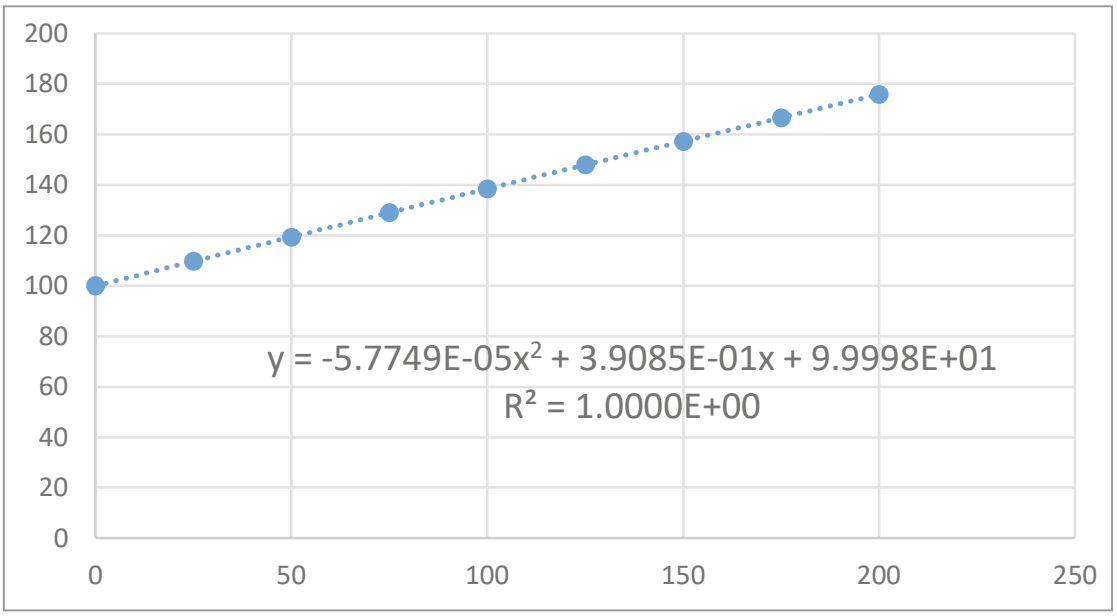
$$\rho = \frac{M}{V} = \frac{6M}{\pi D^3} = \frac{6 \times 1\,000}{3.1416 \times 8^3} = 3.730\ \text{g/cm}^3$$

(a)

$$u_\rho = \sqrt{\left(\frac{\partial \rho}{\partial M} u_M\right)^2 + \left(\frac{\partial \rho}{\partial D} u_D\right)^2}$$

Chapter 7

7.8.1
(a)



$$R(t) = R(0)\ \left[1 + At + B\ t^2\right] = 99.998\ \left(1 + 3.91\text{E-}3\,t\text{–}5.78\text{E-}7\ t^2\right)$$

(b)

(x) Temp. Std. (°C)	(y) Ω	$R(t)$ Ω	$[y-R(t)]^2$
0.00	99.99	99.9980	6.4E-05
25.00	109.74	109.7332	4.6828E-05
50.00	119.40	119.3961	1.4996E-05
75.00	128.99	128.9869	9.5365E-06
100.00	138.50	138.5055	3.036E-05
125.00	147.95	147.9519	3.6936E-06
150.00	157.32	157.3261	3.7792E-05
175.00	166.63	166.6282	3.2874E-06
200.00	175.86	175.8580	3.8416E-06
		Σ	0.000214336
		u_{fit}	0.005977

(c) $\frac{\partial R}{\partial t} = R0\ (A + 2Bt)\ \frac{\Omega}{^\circ\text{C}} = 99.998\ \left(3.91\ \times\ 10^{-3} - 2\ \times\ 5.78\ \times\ 10^{-7} t\right)\ \frac{\Omega}{^\circ\text{C}}$

Chapter 8

8.9.1. (b)
8.9.2. (a)
8.9.3. $y = 910$ bar

$$T_L = 900 \text{ bar} \quad U = 10 \text{ bar} \rightarrow u = 5 \text{ bar}$$

$$P_c = \varphi\left(\frac{y - T_L}{u}\right) = \varphi\left(\frac{910 - 900}{5}\right) = \varphi(2) = 0.9772 \rightarrow 97.72\%$$

8.9.4. $P_c = \varphi\left(, \frac{T_U - y}{u}\right) - \varphi\left(, \frac{T_L - y}{u}\right)$

$$T_U = 34.0\ ^\circ\text{C} \quad T_L = 30\ ^\circ\text{C} \quad u(y) = 0.5\ ^\circ\text{C}$$

$y = 34\ ^\circ\text{C} \quad P_c = \varphi\ \left(\frac{34 - 34}{0.5}\right) - \varphi\ \left(\frac{30 - 34}{0.5}\right) = \varphi(0) - \varphi(-16) = 0.5 - 0$
$= 0.5\ (50\%)$

$y = 32\ ^\circ\text{C} \quad P_c = \varphi\left(\frac{34 - 32}{0.5}\right) - \varphi\left(\frac{30 - 32}{0.5}\right) = \varphi(4) - \varphi(-4) = 1 - 0$
$= 1\ (100\%)$

$y = 33\ ^\circ\text{C} \quad P_c = \varphi\left(\frac{34 - 33}{0.5}\right) - \varphi\left(\frac{30 - 33}{0.5}\right) = \varphi(2) - \varphi(-6) = 0.9772 - 0$
$= 0.9772\ \ (97.72\%)$

8.9.5. (c)
8.9.6. $T_U = 500$ mg/kg $\quad U = 4$ mg $\rightarrow u = 2$ mg $\quad P = 0.99$

$$P = \varphi\left(\frac{y - T_U}{u}\right) = \varphi\left(\frac{y - 500}{2} = z\right) = 0.99 \rightarrow z = 2.33$$

$$2.33 = \left(\frac{y - 500}{2}\right) \rightarrow y = 504{,}66 \cong 505 \text{ mg/kg}$$

8.9.7. (d)

8.9.8.

$$P = \varphi\left(\frac{T_L - y}{u}\right) = \varphi\left(\frac{715 - y}{2.5} = z\right) = 0.999 \rightarrow z = 3.08$$

$$3.08 = \left(\frac{715 - y}{2.5}\right) \rightarrow y = 707 \text{ kg/m}^3$$

8.9.9.

(a) (83 ± 3) ° C

(b) TUR $= TL/U \rightarrow U = TL/\text{TUR} = 3$ ° C$/3 = 1$ ° C

8.9.10.

(a) $T_L = 44.0$ ° C and $T_U = 50.0$ ° C

(b) TUR $= TL/U = 5 \rightarrow U = TL/\text{TUR} = 3$ ° C$/5 = 0.6$ ° C

$w = 1.5$ $U = 0.9$ ° C $\rightarrow$ $A_L = T_L + w = 44.9$ ° C and $A_U = T_U - w = 49.1$ ° C

Chapter 9

9.5.1

(R)	The calibration certificate is a technical record that enables the user to measure instrument conformity assessment
(W)	The laboratory that performs the measuring instrument calibration will never make any recommendation on the periodicity of calibration
(R)	The calibration laboratory does not usually adjust the measurement instrument; however, results should be reported before and after adjustment if this is done
(R)	The certificate must present calibration results in the units of measure used and declare the measurement uncertainty, scope factor, and confidence level used
(W)	The environmental conditions of the calibration site only need to be declared, if they affect the calibration result